

## RND-6: SAFE IN-TIME COVARIATES ??? CUPAC-STYLE ????????

> ??????: ? ??????. ??????????? ?????????????? ????????? ? ??????? ???????  
> A/B ? CUPAC A ? A+B ? A+B+C; ?????????? ?????????? ?????????? open-source ?????????? ?  
> ?????????? ?????????????? ?????????? ? delta-?????????? ? A-only (?????? «Expanded public dataset  
> validation»). ?????????? ??????? ? B/C-???????? ? notebook.ipynb (??????????, seed=11).

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??????????, ??? ? ? ?????????????????? ??????????? ??????????? in-time ?????????? ??????  
scikit-CUPAC ?????????????????? ?????????? ?????????? (? ??????????? ?????????? ?????????) ???  
?????????? ?????????????????? ?????????????? ??????? ATE.

?????? ?????????? (SAFETY LAYER)

? **A** pre-treatment ? ??? ? CUPAC;  
? **B** expert-safe in-time ? ?????????? second-stage ??????????;  
? **C** balance-gated in-time ? ?????????? ?????????? ????? balance/missingness gate;  
? **D** dag-required ? ??? estimator (????? causal-????????);  
? **E** mediator-risk, **F** leakage ? ?????????? ??? ?????????? ?????? ATE;  
? **unsafe\_demo** ? ?????? ?????????????? ??????, ? ??????????

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ab\_hypex (A/B baseline) • hypex\_cupac (reference/parity, ??? ??????) •  
sklearn\_cupac\_A ? sklearn\_cupac\_A\_plus\_B\_linear ? sklearn\_cupac\_A\_plus\_B\_plus\_C\_linear  
(????????? ?????????? ??????????) • unsafe\_demo\_optional (?????????????).

?????????? ? ? ?????????? (N=8000, SEED=11, ?????????? ATE = 2.8)

| method                               | ate   | p_value | variance_reduction_vs_ab_% | incremental_vs_predecessor_% | safety_status  |
|--------------------------------------|-------|---------|----------------------------|------------------------------|----------------|
| ab_hypex                             | 2.758 | 0.0     | 0.0                        | ?                            | ok             |
| hypex_cupac                          | 2.785 | 0.0     | 13.56                      | ?                            | reference_only |
| sklearn_cupac_A                      | 2.801 | 0.0     | 25.63                      | 25.63                        | ok             |
| sklearn_cupac_A_plus_B_linear        | 2.775 | 0.0     | 42.38                      | 22.52                        | ok             |
| sklearn_cupac_A_plus_B_plus_C_linear | 2.766 | 0.0     | 52.80                      | 18.07                        | ok             |
| unsafe_demo_optional                 | 1.000 | 0.0     | 83.17                      | ?                            | unsafe_demo    |

(????????? ?????????? ????????? ? ?????? ?????????????? ?????????? ? ? ?????????? ??????????)

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- ?????? ?????????????? ??????? ???? ?????????????????? ?????????: A (~26%) ? +B (~42%) ? +B+C (~53%)  
????????? ?????????? ?????????????? A/B, ??? ??? ATE ?????????? ?????????????? (? 2.8).
- CUPAC parity: hypex\_cupac (~14%) ?????????? ? ? ?????? ? sklearn\_cupac\_A;  
????????? ?????????? ????????? ? R&D ? ?????????????? ??? ?????????.
- Balance/missingness gate ?????????? A/B/C (|SMD|>0) ? ?????????? D/E/F ? unsafe\_demo  
(|SMD| ? 0.45?1.14, p>0) ? ?????????, ?????? ? ? ?????? ?????? ? estimator.

4. unsafe\_demo ???? ???????????? «????????? ??????????» (~83%), ?? ??????? ?????? ATE  
?????? ?? ????????? 2.8 ? ????????? ? 1.0 (?. ??????) ? ?????????? outcome-derived  
?????????; ? ?????????? ?????? ?? ??????.

????????????? ? ????????????

? balance-test ? ????????????? gate (????? ?????????? ?????????? E/F), ? \*\*??\*\* ??????????????  
????????? ?????????????; ??? D-????????? ?????? ??????? DAG/causal-?????.  
? E/F ??????????; safe-intime ????????????? \*\*????\*\* ? ????????? second-stage, ?? ?  
???????????? ML-??????.

?????? ?????????? ?? ?????????? ??????

? \*\*????????\*\* ????????????? ?????? ??????? \*\*A+B+C\*\* (?????? ????): ?????? ??? ??????  
A/B/C/D/E/F ??????? ?? ??????????  
? \*\*????????? ?????????? ?????????? ?????? ?????????????? ? ?????????? A-only\*\* CUPAC-?????????  
????? ????? (download + schema\_summary) ?????? Orange Belgium, Lenta, Criteo, X5  
RetailHero,  
MegaFon, Lazada/DESCN, CUVET-policy, MT-LIFT, Dunnhumby ? semi-synthetic causal-?????????  
? \*\*Hillstrom\*\* ?????????? pipeline/A-only, \*\*??\*\* B/C.  
? ????????? safe in-time \*\*B/C\*\* ??????????: ?????????? (f\*, 'PC\*', 'X\_\*'),  
?????  
??? ???-????????? ?????????? (A), ??? ???-treatment ?????????? (E/F). ?????????? ? ??????????  
????????????? ??? ?????????? ??? ?????????? (X5), ??? ?? ?????????????????? (Dunnhumby).  
? \*\*?????? A+B+C-????????? ? ?????????? ?????????? ?????????? ? ?????????????\*\* ? ?????????  
????????????????, ??? ?????? ??? ? ?????? ?????????? B/C-????????? ? timing-evidence.  
? ?????? ? ????????????? ?????????????? ??????????: \*\*docs/06\_real\_dataset\_raw\_audit.md\*\*.

## EXPANDED PUBLIC DATASET VALIDATION

????????????? ?????????? (????? ??? ?????????????? uplift-????????? + ?????????? ??????????/????????? +  
????????? ?????????? + ?????????????? delta-????????? ?????????). ?????? ?????? ? ??????  
?????????????????????: docs/06\_expanded\_dataset\_scouting.md.

????????????????????? ?????????? ?????????? (?? ?????? ???, ?? ?? ??????????????):  
? \*\*A\*\* (pre-treatment) ? \*\*9+\*\* ?????????? ?????????? ?????? A-????????? ?? ?????? (+  
?????????), ??  
????????????? delta-????????? ?????????? ? ?????????????? ?? 5 ?? ??? (hillstrom, orange\_belgium,  
lazada\_descn, lenta, x5 ? ?????? ???). ?????????? A/sandbox-????????? ?????????? ?? ??????  
??????/????????????????? (criteo ? access\_blocked; open\_bandit/dunnhumby/criteo\_private\_ad ?  
sandbox,  
?? ?????????? ?? ?????????? A-?????????).  
? \*\*B\*\* (expert-safe in-time) ? \*\*0\*\*?: ?? ??? ?????????? ?????? ?? ?????????????? ?????????????,  
????????????? ?????????????????? in-time ??????????-????????????? ?????????.  
? \*\*C-????????\*\* (????????, ?????????) ? \*\*0\*\*; sandbox-????????????? C ?????????????? ?????????  
?? 2 event-log ?????????? (Open Bandit, dunnhumby), ?? ??? ?? ?????????? ??????????  
? \*\*D-????????\*\* ? \*\*5\*\* (Criteo 'exposure'; Lazada biased-train propensity; Open Bandit  
action/position/pscore; CriteoPrivateAd display/privacy-????????; dunnhumby  
campaign/coupon).  
? \*\*E/F/unsafe-????????\*\* ? \*\*7+\*\* (Hillstrom 'visit'/'conversion'; Criteo 'exposure';  
X5 ???-????????????????? ?????????; Open Bandit reward-????????????; CriteoPrivateAd delayed  
click/sale; dunnhumby redemption ? ??).

?????-A ????????? (real RCT/uplift): Hillstrom, Lenta, X5 RetailHero, Orange Belgium, Lazada/DESCN (???????????????? test), Criteo (????? A-only).  
C/D-«????????» (?? A/B): Open Bandit (logged bandit), dunnhumby Complete Journey (observational), CriteoPrivateAd (advertising logs) ? ?????? ??? \*????????????\*  
C/D-????????,  
?? ?????????? causal-????????????? safe in-time.

> ?????? A+B+C-????????? ?? ?????????? ?????????? ?????????? ?? ??????????????. ??????  
> C/D-????????????????????? ?????? ?????????????? ?? event-log / bandit / advertising / journey  
> datasets, ?????????? ?? ?????????? ?????????? A/B uplift benchmark.

Delta-????????? ?? ?????????? ?????? (A-only ?????????? ab\_hypex ? sklearn\_cupac\_A, seed=11).  
????????? ? ?????? ?????????????? python tools/audit\_datasets.py --delta; ?????????????? ?????????????? ?  
expanded\_dataset\_delta\_results.csv, ?????????? ? figures/.

| dataset        | validation_level | n       | ATE (A/B) | var.reduction | CUPAC-A vs A/B, % | A-???????????? | safety |
|----------------|------------------|---------|-----------|---------------|-------------------|----------------|--------|
| hillstrom      | A_only_real      | 64 000  | 0.597     | 0.06          | 18                | ok             |        |
| orange_belgium | A_only_real      | 11 896  | ?0.0028   | 6.06          | 336               | ok             |        |
| lazada_descn   | A_only_real      | 181 669 | 0.0037    | 5.41          | 83                | ok             |        |
| lenta          | A_only_real      | 687 029 | 0.0075    | 14.36         | 196               | ok             |        |
| x5_retailhero  | A_only_real      | 200 039 | 0.0332    | 16.10         | 10                | ok             |        |

? \*\*CUPAC-A ???? ?????????? ?????????? ?????????? 0?16%\*\* ?? ?????????? ?????????; ?????????? ?????? ???,  
??? ?????? ?????????????????? pre-treatment ?????????? ? lenta (????????????? ???-????????? ??????????,  
~14%) ? X5 (engineered ?????????? ??????????, ~16%). ATE ?????????? ?????????????? (????????????????).  
? \*\*hillstrom\*\* ? pipeline/A-only smoke: ?????????? 'spend' ?????????? ?????????, A-????????? ??????  
????????? (~0.06%).  
? \*\*????????? '+B' ? '+B+C' ?? ?????????? ?????????? ?? ?????????????????\*\*?: ?????????? B/C-????????? ???,  
?????????  
????????????????? delta-????????? B/C ??-????????? ?????????????????? ?????????? ?? ?????????? (?????????  
?????).  
? \*\*?? ?????? ? delta-?????????:\*\* 'criteo' ? ?????????? ????????? HTTP 403 (access\_blocked);  
open\_bandit ? ?????? obp ?????????????? ? pandas 2.x (DataFrame.drop("col", 1)),  
runtime-blocked sandbox; dunnhumby / criteo\_private\_ad ? ?????????? ?????????? ??????????????????  
registration-gated ?????????? ? ??? safe B/C ?? ?????????????????? (????????? ?????????? ??????????  
?????????).

????????????????? ?? ?????????? ? HYPEX

????????? ?????????????? ?????????????? ?????????? ?? ?????????????? ? ?????????????? ? HypEx CUPAC; ?? ??????????  
????????????? ?????????? ?????????????? A-only ?????? CUPAC-A (0?16% ?????????? ?????????????, 5 ?????????????).  
????????? ?????????????? ? HypEx ??????????????:  
1. ?????????????????? ?????? B/C ?? ?????????????? ?????????????????? ?????????, ? ?? ?? ??????????????. ??????????????  
????????????? ?????????, ??? ?????????????? ?????????????? ?? ?????????????? B/C (????????????? ?????????????????? safe  
in-time ?????????? ? timing-evidence ???; ?? «Expanded public dataset validation»). ??????????  
B/C-????????????????? ?????????? ?????????????????? ?? ?????????????????? ????????? (VarWar-?????????), ??? timing ?  
?????????????  
????????????? ?????????????? ? ???????????.  
2. ?????????????????????? balance/missingness gate ??? ?????????????????????? ??????????????.

3. ?????????????? ?????? E/F ? ?????????? ?????? unsafe\_demo.

????????????? ?????????????? ?????????????? CUPAC ?? ?????????? ?????? ? VarWar (????????? ?????????? ~34% ? ~33%; ?????? ??????????, ?? ??????????????) ? ??? ? ?????? ?????? ?????????????? ??????????, ?? ?????????  
?????? ??????????? B/C.

???????????????????

bash  
pip install -e .[ml]  
??. NOTEBOOK.IPYNB; ?????? ATE–CI ? ?????????? ?????????????? ?? RND\_REPORTS.BENCHMARK

DELTA-????????? ?? ?????????? ??????????? (????????? GITIGNORED DATA/, RAW ?? ??????????????):

python tools/audit\_datasets.py --delta # ? expanded\_dataset\_delta\_results.csv + figures/  
python tools/generate\_pdf.py # ?????????????? report.pdf